

UNDERSTANDING WIND DYNAMICS IN VR: VISUAL ANALYTICS FOR TERRAIN AND ARCHITECTURE

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INTRODUCTION

Wind plays a significant role across meteorology, architecture, urban planning, and energy. It influences weather patterns, storm trajectories, and atmospheric circulation, and helps guide architectural design for natural ventilation, wind load reduction, and occupant comfort. In cities, wind behaviour affects pedestrian comfort, heat island mitigation, and building layouts, while optimized wind flows can enhance on-site power generation. Traditional 2D plots only partially capture wind's vertical dynamics. VR-based 3D visualization, by contrast, embeds wind layers within realistic terrain and supports intuitive, immersive exploration. Our VR application delivers **interactive visual analytics**, enabling users to **explore wind patterns** across **space** and **time** at **high frame rates (≥ 90 FPS)** to drive more **informed decision-making**.

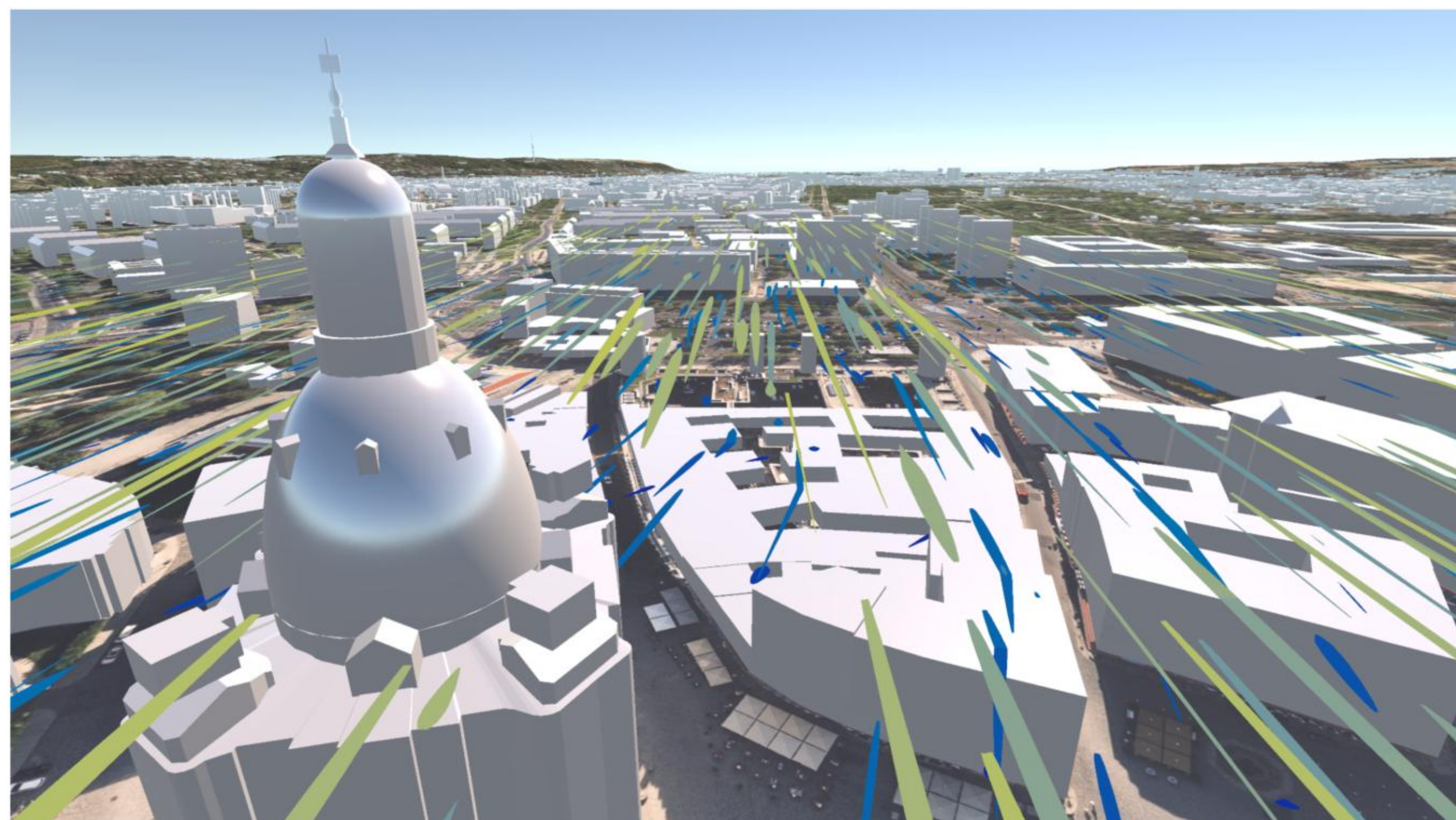


Fig 1: (Top) Bird's Eye View: Visualization of wind data of 19th January 2007 as streamlines in Dresden. (Bottom) The color scale used for coloring the streamlines according to their speed. Left corresponds to 0 (m/s) to right 32.7 (m/s)

EVALUATION

Our evaluation includes:

- Performance test on high end PC with RTX4070Ti and MetaQuest3 at 120Hz. The application consistently delivered between 116–126 FPS with a mean of **109 FPS** during **full VR navigation and dynamic data interactions**.
- A 45 minutes WebEx based expert interview incorporating
 - Two wind and climate modeling professionals (one meteorologist and one domain expert in regional climate analysis and monitoring) from Saxon State Ministry from Environment and Agriculture
 - Explored freely and identified wind patterns** as part of their task
 - Feedback was captured live and summarized post-session which highlighted that wind patterns appeared plausible but pointed out limitations such as missing vertical turbulence (due to dataset limitations), difficulty comparing time steps, and a desire for constant-streamline mode.
 - Both participants liked the prototype and **agreed** that it can serve as a **powerful tool, offering visual analytics for understanding wind dynamics**.

VISUAL ANALYTICS AND METHODOLOGY

Supporting interactive VR exploration of wind over a 9TB dataset posed two main challenges:

- enabling low-latency navigation across large spatial regions
- rendering real-time, terrain-aligned flow visualizations while maintaining ≥ 100 FPS.

We implemented

- Tile-based streaming loads only the central ~ 1 km² terrain-wind tile and its neighbours, keeping memory use minimal and enabling sub-second viewport updates.
- Wind data is terrain-integrated. Wind vectors are vertically shifted and slope-aligned using elevation info to respect actual topography.
- Animated GPU-powered streamlines visualize airflow intuitively by flowing particles through dynamically-generated 3D vector field textures.
- Built on Unity 2022.3 URP, the system integrates XR Interaction Toolkit, Cesium for Unity, and VFX Graph, rendering wind dynamics over realistic terrain with buildings and surface textures.
- Users roam Saxony freely using teleportation and joystick navigation, with on-controller UI for view switching and smooth, real-time interaction.

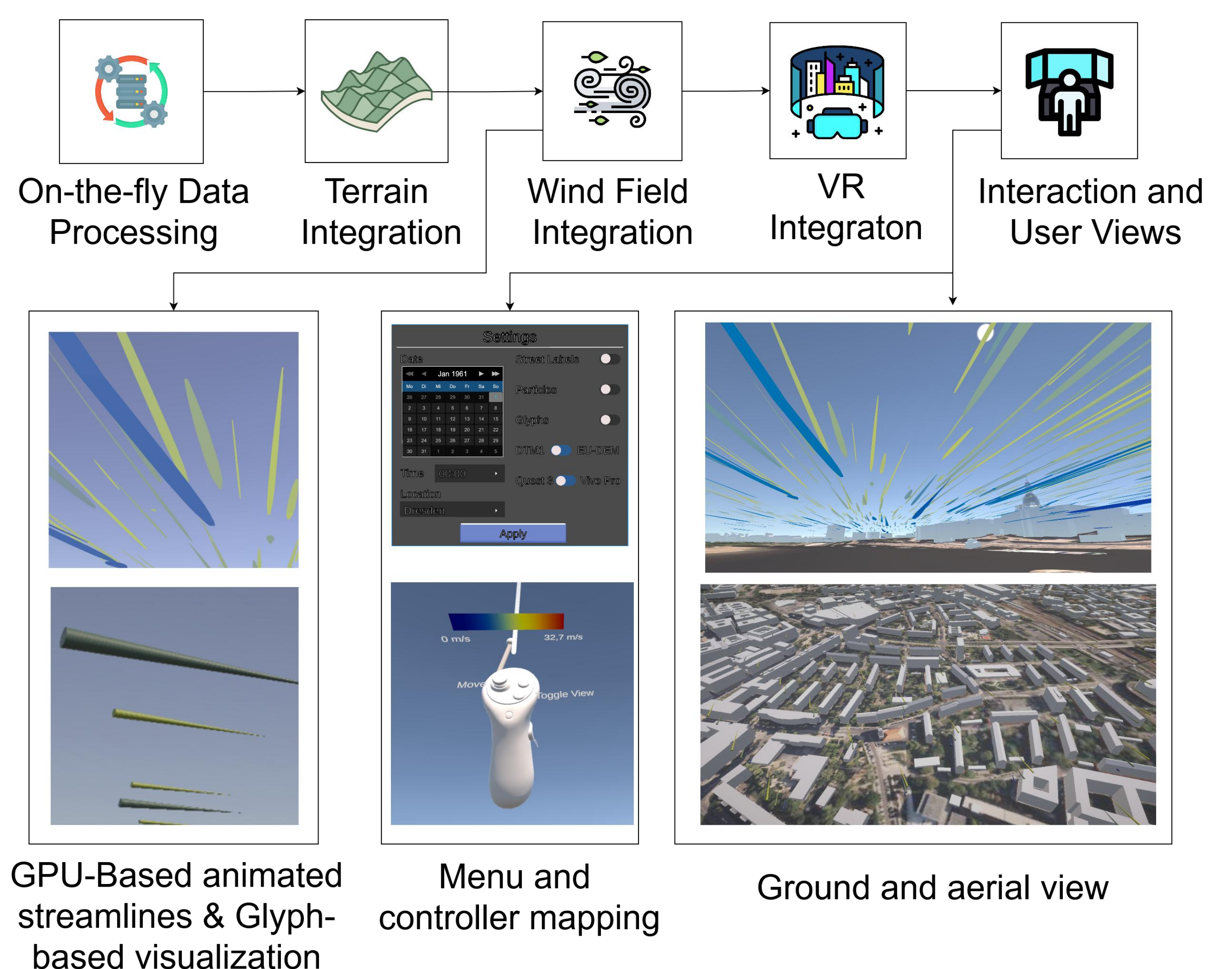


Fig 2: Showcasing wind visualization methodology and visual analytics

KEY TAKEAWAY

- This work addresses **real-time GPU-based particle generation, on-demand terrain and wind tile loading** and **latency-aware UX design** which have not been handled in previous work.
- The framework is data-agnostic and works with any dataset that provides gridded vector wind data (speed and direction) across temporal slices, especially in NetCDF format. Adapting to other regions requires:
 - Equivalent vector wind file format,
 - Mapping to terrain-relative heights, and
 - Terrain/building data can be integrated using Cesium's APIs without requiring use of Cesium World Terrain or OSM buildings.



<https://scads.ai/>

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FUNDED BY



Bundesministerium
für Forschung, Technologie
und Raumfahrt



SACHSEN Diese Maßnahme wird gefördert durch die Bundesregierung aufgrund eines Beschlusses des Deutschen Bundestages. Diese Maßnahme wird mitfinanziert durch Steuermittel auf der Grundlage des von den Abgeordneten des Sächsischen Landtags beschlossenen Haushaltes.